

Feature Specification: Polyglot Language Memory Game

Goal

Implement a language-learning memory game that combines the classic card-matching mechanic with multilingual vocabulary training.

The game shall help users memorize vocabulary across multiple languages simultaneously by forcing active recall and cross-language associations.

Concept

Instead of matching identical images or symbols, users match words that represent the same concept in different languages.

Example:

Language	Word
German	Haus
English	house
Spanish	casa
French	maison
Swedish	hus
Polish	dom

All words belong to the same semantic concept:

```
{
  "concept_id": "house",
  "de": "Haus",
  "en": "house",
  "es": "casa",
  "fr": "maison",
  "sv": "hus",
  "pl": "dom"
}
```

Game Modes

Mode 1: Pair Matching

User must find two matching words.

Example:

house ↔ Haus
casa ↔ Haus
maison ↔ house

A match occurs when both words belong to the same concept.

Mode 2: Triple Matching

Three cards must be matched.

Example:

house
casa
maison

All belong to the same concept.

Mode 3: Polyglot Matching

Advanced mode.

User must find all available language variants of a concept.

Example:

house
Haus
casa
maison
hus
dom

All six cards must be collected.

Adaptive Difficulty

Difficulty is dynamically adjusted based on:

- success rate
- response time
- previous mistakes
- memory strength score

Difficulty Levels

Beginner

- 4 concepts
- 8 cards
- 2 languages

Intermediate

- 8 concepts
- 16 cards
- 3–4 languages

Advanced

- 12 concepts
- 24 cards
- 5–6 languages

Expert

- 20+ concepts
- 40+ cards
- all enabled languages

Personalized Learning

The game shall prioritize vocabulary that:

- has low memory strength
- was answered incorrectly recently
- has not been reviewed for a long time
- is part of the current learning goal

Selection priority:

Weak words > Forgotten words > New words > Known words

Gameplay Rules

Card Flip

User clicks a card.

Card reveals:

- word
- language indicator

Example:

house 

Correct Match

If cards belong to same concept:

- cards stay visible
 - score increases
 - memory strength increases
-

Incorrect Match

If cards do not belong together:

- short animation
- cards flip back
- mistake is stored

Mistake example:

```
{
  "user_id": 123,
  "word_a": "maison",
  "word_b": "dom",
  "result": "incorrect"
}
```

Interference Detection

Very important.

Track language confusion patterns.

Example:

User repeatedly confuses:

```
maison  
dom
```

System stores:

```
{  
  "confusion_pair": [  
    "maison",  
    "dom"  
  ],  
  "count": 5  
}
```

These pairs shall appear more often in future games.

Scoring

Base score:

```
Correct match = 100 points
```

Speed bonus:

```
< 3 seconds = +50  
< 2 seconds = +100
```

Streak bonus:

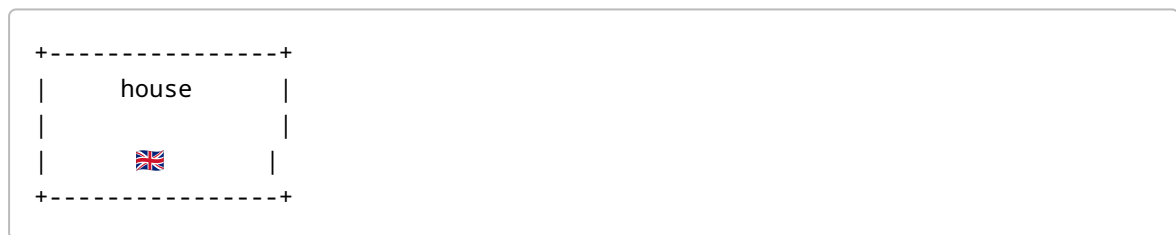
```
5 matches = +200  
10 matches = +500
```

Visual Design

Design goals:

- clean
- distraction-free
- language-focused
- mobile-friendly

Card should display:



No unnecessary graphics.

Vocabulary must remain the focus.

AI Integration

The AI system shall:

1. Select optimal concepts.
 2. Generate example sentences.
 3. Generate images for difficult concepts.
 4. Identify confusion patterns.
 5. Adapt future sessions.
-

Post-Game Review

After every game:

Show:

- accuracy
- average response time
- difficult words
- confusion pairs

Example:

Most difficult concepts:

house
pump
excavator

Common confusions:

maison ↔ dom
pump ↔ bomba

Future Extensions

- Audio Memory (hear word, find translation)
- Image Memory (image ↔ word)
- Sentence Memory
- Verb Conjugation Memory
- Technical Vocabulary Packs
- Multiplayer Challenges
- Daily AI-generated Memory Boards

Technical Requirements

Frontend:

- React
- TypeScript
- TailwindCSS

Backend:

- FastAPI

Database:

- PostgreSQL

State Management:

- Zustand

Game Engine Requirements:

- <100 ms card response
- mobile support
- keyboard support

- offline-capable cache

Code Quality:

- strict TypeScript
- reusable components
- unit tests
- E2E tests
- accessibility compliant (WCAG AA)